IBM Skills Build PBL Program

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Report

Title: AI-Powered Personalized Learning Platform for Underprivileged Students

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Introduction:

Access to quality education remains a major challenge for underprivileged communities. Traditional educational methods often fail to address individual learning needs, leading to disengagement, poor performance, and high dropout rates. With advancements in Artificial Intelligence (AI), we now have the opportunity to build systems that personalize education, adapting content and assessments based on each learner's unique abilities and pace.

This project report outlines the development of an **AI-powered personalized learning platform** designed to improve educational outcomes—especially for underprivileged students—by providing tailored content, interactive assessments, and adaptive learning paths using the Gemini API from Google.

Problem Statement:

Despite the availability of online education resources, many students—especially those from underprivileged backgrounds—struggle due to lack of personalization, motivation, and contextual relevance in learning. The need exists for a system that:

- Understands the learner's level,
- Adapts to their pace,
- Provides personalized guidance and assessments,
- Bridges knowledge gaps effectively.

Our project addresses this need by creating a customizable, AI-driven educational assistant.

Objective

To design and implement an intelligent tutoring system using Google's Gemini AI that:

- Generates topic-wise questions and answers tailored to different cognitive levels (MCQ, descriptive, etc.),
- Delivers interactive and meaningful learning sessions,
- Improves comprehension, retention, and educational equity.

Why This Problem?

The UN's Sustainable Development Goal 4 (Quality Education) emphasizes inclusive and equitable education for all. However, marginalized communities often lack access to teachers, personalized mentoring, and structured content. By automating key teaching and learning functions, our solution democratizes access to quality education and nurtures independent learners.

Solution Overview

Our solution is an AI-powered personalized learning platform that adapts dynamically to learners' inputs, offering:

Key Features

- **Subject-wise Smart Q&A Generator**
  Users input a subject and a question type (MCQ, Explain, One sentence, Descriptive). The system generates tailored content using Gemini AI.
- **Adaptive Learning Paths**
  Based on learner inputs, future iterations will recommend topics and assessments suited to learner performance.
- **Multilingual & Inclusive UI (Future Scope)**
  Designed to be inclusive for regional language support and accessible on low-resource devices.

Technical Implementation

Tools Used

- **Google Gemini API**: For AI-driven content generation
- **IPyWidgets & Colab**: For interactive UI prototyping
- **Python**: Core scripting language

Why Gemini and Google AI Tools?

- **Generative Intelligence**: Gemini's latest models provide accurate, coherent, and context-aware educational content.
- **Scalability**: Easily integrable into web or mobile platforms.
- **Ease of Use**: IPyWidgets offers fast prototyping and testing with minimal UI setup.
- **Future Extensibility**: Gemini APIs support multimodal capabilities that can evolve with the platform.

Expected Outcomes

- Increased engagement and comprehension among learners.
- Improved accessibility to quality education in resource-constrained areas.
- A scalable foundation for a complete AI learning ecosystem.

Conclusion

The platform aims to bridge educational gaps using AI by offering adaptive, intelligent learning experiences tailored to the learner. By leveraging the power of Google’s Gemini AI, we envision a future where every student regardless of background can access quality education, learn at their pace, and thrive in their academic journey.